

Q1 2021 (01 Sep Shift 2)

An engine is attached to a wagon through a shock absorber of length 1.5 m. The system with a total mass of 40,000 kg is moving with a speed of 72 kmh^{-1} when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0 m. If 90% of energy of the wagon is lost due to friction, the spring constant is $\dots \times 10^5 \text{ N/m}$

Q2 2021 (01 Sep Shift 2)

A body of mass ' m ' dropped from a height ' h ' reaches the ground with a speed of $0.8\sqrt{gh}$. The value of workdone by the air-friction is :

- (1) $-0.68mgh$
- (2) mgh
- (3) $1.64mgh$
- (4) $0.64mgh$

Q3 2021 (31 Aug Shift 2)

A block moving horizontally on a smooth surface with a speed of 40 m/s splits into two parts with masses in the ratio of 1 : 2. If the smaller part moves at 60 m/s in the same direction, then the fractional change in kinetic energy is :-

- (1) $\frac{1}{3}$
- (2) $\frac{2}{3}$
- (3) $\frac{1}{8}$
- (4) $\frac{1}{4}$

Q4 2021 (31 Aug Shift 1)

A block moving horizontally on a smooth surface with a speed of 40 ms^{-1} splits into two equal parts. If one of the parts moves at 60 ms^{-1} in the same direction, then the fractional change in the kinetic energy will be $x : 4$ where $x = \dots$

Q5 2021 (27 Aug Shift 2)

The height of victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?

[Given $1\text{ cal} = 4.2\text{ J}$ and specific heat of water $= 1\text{ cal g}^{-1}\text{ }^{\circ}\text{C}^{-1}$]

(1) 0.147°C

(2) 14.76°C

(3) 1.476°

(4) 0.014°C

Q6 2021 (27 Aug Shift 1)

Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angle 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is

Q7 2021 (26 Aug Shift 1)

A uniform chain of length 3 meter and mass 3 kg overhangs a smooth table with 2 meter laying on the table. If k is the kinetic energy of the chain in joule as it completely slips off the table, then the value of k is

(Take $g = 10\text{ m/s}^2$)

Answer Key

Q1 (16)

Q2 (1)

Q3 (3)

Q4 (1)

Q5 (1)

Q6 (2)

Q7 (40.00)

Q1 (16)

$$\text{Work} = \Delta K, E$$

$$W_{\text{friction}} + W_{\text{spring}} = 0 - \frac{1}{2}mv^2$$

$$-\frac{90}{100} \left(\frac{1}{2}mv^2 \right) + W_{\text{Spring}} = -\frac{1}{2}mv^2$$

$$W_{\text{spring}} = -\frac{10}{100} \times \frac{1}{2}mv^2$$

$$-\frac{1}{2}kx^2 = -\frac{1}{20}mv^2$$

$$\Rightarrow k = \frac{40000 \times (20)^2}{10 \times (1)^2} = 16 \times 10^5$$

Q2 (1)

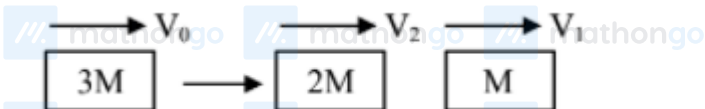
Work done = Change in kinetic energy

$$W_{\text{mg}} + W_{\text{air-friction}} = \frac{1}{2} m(.8\sqrt{gh})^2 - \frac{1}{2} m(0)^2$$

$$W_{\text{air-friction}} = \frac{.64}{2} mgh - mgh = -0.68mgh$$

Option (1)

Q3 (3)



$$3MV_0 = 2MV_2 + MV_1$$

$$3V_0 = 2V_2 + V_1$$

$$120 = 2V_2 + 60 \Rightarrow V_2 = 30 \text{ m/s}$$

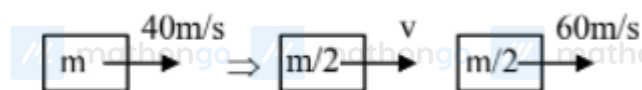
$$\frac{\Delta K.E.}{K.E.} = \frac{\frac{1}{2}MV_1^2 + \frac{1}{2}2MV_2^2 - \frac{1}{2}3MV_0^2}{\frac{1}{2}3MV_0^2}$$

$$= \frac{V_1^2 + 2V_2^2 - 3V_0^2}{3V_0^2}$$

$$= \frac{3600 + 1800 + 4800}{4800}$$

$$= \frac{1}{8}$$

Q4 (1)



$$P_i = P_f$$

$$m \times 40 = \frac{m}{2} \times v + \frac{m}{2} \times 60$$

$$40 = \frac{v}{2} + 30$$

$$\Rightarrow v = 20$$

$$(\text{K.E.})_1 = \frac{1}{2} m \times (40)^2 = 800 \text{ m}$$

$$(\text{K.E.})_f = \frac{1}{2} \frac{m}{2} \cdot (20)^2 + \frac{1}{2} \cdot \frac{m}{2} (60)^2 = 1000 \text{ m}$$

$$|\Delta \text{K.E.}| = |1000 \text{ m} - 800 \text{ m}| = 200 \text{ m}$$

$$\frac{\Delta \text{K.E.}}{(\text{K.E.})_i} = \frac{200 \text{ m}}{800 \text{ m}} = \frac{1}{4} = \frac{x}{4}$$

$$x = 1$$

Q5 (1)

Change in P.E. = Heat energy

$$mgh = mS\Delta T$$

$$\Delta T = \frac{gh}{S}$$

$$= \frac{10 \times 63}{4200 \text{ J/kgC}}$$

$$= 0.147^\circ \text{C}$$

Q6 (2)

Given $W_A = W_B$

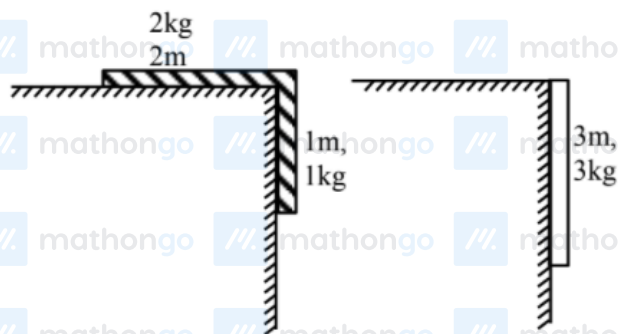
$$F_A d \cos 45^\circ = F_B d \cos 60^\circ$$

$$F_A \times \frac{1}{\sqrt{2}} = F_B \times \frac{1}{2}$$

$$\frac{F_A}{F_B} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$x = 2$$

Q7 (40.00)



From energy conservation

$$K_i + U_i = k_f + U_f$$

$$0 + \left(-1 \times 10 \times \frac{1}{2}\right) = k_f + \left(-3 \times 10 \times \frac{3}{2}\right)$$

$$-5 = k_f - 45$$

$$k_f = 40 \text{ J}$$

Ans. 40.00